



Tribhuvan University
Faculty of Humanities and Social Sciences
Ratna Rajyalaxmi Campus
Pre-Board Examination 2025

Bachelor in Computer Application (BCA)

Course Title: Scripting Language

Code No: CACS 254

Semester: 4th

Full Marks: 60

Pass Marks: 24

Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible

Group B

Attempt any FOUR questions.

[4×5= 30]

2. What is dialog box? Explain different dialog boxes with suitable example in JavaScript [5]
3. Discuss about different ways to define functions in JavaScript. Write a complete JS program to demonstrate how arrow function is used. [3 + 2]
4. What are session and cookies? Explain how session is used in PHP with suitable example to demonstrate the different steps in session handling. [2+3]
5. Describe how for loop and foreach loop works with array in PHP with suitable example.
6. What is Ajax? Create a web page that allows user to enter two numbers and displays the multiplication table of the bigger number using Ajax with PHP. [5]
7. What is the importance of server side validation? Create a form to take name and age values from user and validate on server side using PHP. [2 +3]
8. What is CMS? Describe major features of CMS. [5]

Group C

Attempt any ONE questions

[1×10 = 10]

9. Write a program to display all records of student in a web page in tabular format after reading from a database table using PHP (Assume the table contains some records of students). Also include a button for each record in HTML page and include the functionality to delete the corresponding record from database when the button is clicked. [10]
Assume your own database, database connectivity constrains and schema.
10. Discuss about exception handling in PHP and necessary keywords with suitable examples. Explain about abstract class and interface in PHP with suitable examples for each. [5 + 5]
11. Explain about event handling in JavaScript with any two techniques to register event handlers and also list five common events with corresponding handler with the information about their occurrence. Create a signup form for an application with JavaScript validation. The error should appear next to each form element if exists. [5 + 5]



Tribhuvan University
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Bachelor in Computer Applications

Course Title: Numerical Methods

Code No: CACS252

Semester: IV

Full Marks: 60-40

Pass Marks: 24-16

Time: 3 2 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any FOUR questions

[4 × 5 = 20]

2. Applying Jacobi's iteration method to solve the system of equations $11x + y - z = 11$,
 $x + 5y + z = 28$, $-x + y + 8z = 35$ correct to two decimal places
3. Applying Newton's divided interpolation formula, find $y(2.5)$ from the following data

$x :$	1	3	4
$y :$	5	9	11

4. Applying backward interpolation formula to find $y(7.5)$

$x :$	2	4	6	8
$y :$	10	14	15	18

5. Fitting in the straight line $y = ax + b$ from the following data

x	4	6	8	10	12
y	11.1	8.7	6.4	4.7	2.6

6. Find the Eigenvalues of $A = \begin{bmatrix} 4 & 1 & 0 \\ 2 & 0 & 5 \\ 1 & 2 & 0 \end{bmatrix}$ applying power method

7. Solve the non-linear equation $x^3 - 2x - 5 = 0$ using False-position method with initial guesses $x_1 = 2$ and $x_2 = 3$

Group C

Attempt any one question

[10 × 1 = 10]

8. Solve $y' = xy + y^2$ with $y(0) = 1$ for $x = 0.2$ and 0.4 using Runge-Kutta fourth order methods
9. Solve the boundary value problem $y'' + xy' + y = \sin x$, $y(0) = 1$, $y(1) = 2$, dividing the interval $[0, 1]$ into four subinterval
10. Solve the Poisson equation $u_{xx} + u_{yy} = 2x^2y^2$ on the square mesh on the boundary $u = 0$ in the interval $0 < x < 1$, $0 < y < 1$, $h = 1/3$

$$x - 3y - 5z = -19, \quad 2x - y + 3z = 9$$



Tribhuvan University
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Pre-Board Examination 2026

Bachelor in Computer Application (BCA)

Course Title: Operating System

Code No: CACS 251

Semester: V

Full Marks: 40

Pass Marks: 16

Time: 2 hours

Group B

Attempt any four questions. [4 x 5= 20].

2. Explain different types of kernels.
3. What is Critical Section problem? Explain in details.
4. Given the following page references as
0,9,0,1,8,1,8,7,1,2,8,2,7,8,2,3,8,3.
Calculate page fault for LRU, FIFO page replacement algorithm.
5. Compare the bitmap and linked list implementation of disk free-space management. How much space required in memory to store bitmap for 20 GB hard disk with 2KB block size.
6. Define the term authorization and authentication. Explain in detail about Cryptography with encryption and decryption.
7. What is memory fragmentation? Explain Coalescing and Compaction with suitable example.
8. Explain the condition for occurrence of dead lock in the system. Explain Bankers algorithm with suitable example.

Group C

Attempt any one questions. [1 x 10= 10]

9. List out various conditions for deadlock with its definition. What do you understand by deadlock detection and recovery? Discuss banker's algorithm with suitable example.
10. Can we prevent our system from race condition? Calculate Average Waiting and Average Turnaround time of the given set of processes in table below using SJF, SRTF and RR scheduling algorithm. [Note: Quantum Time for RR=2]

Process id	Arrival Time	Execution Time
A	0	6
B	2	12
C	6	19
D	15	5
E	23	5
F	23	8

11. Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130.
Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk scheduling algorithms?
a. FCFS b. SSTF c. SCAN d. LOOK e. C-SCAN



Tribhuvan University
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Pre-Board Examination 2026

Class R.N.

Bachelor in Computer Application (BCA)

Course Title: Database Management System

Code No: CACS 255

Semester: IV

Full Marks: 40

Pass Marks: 16

Time: 2 hours

Group B

Attempt any FOUR [4*5=20]

2. Define Database. What are the advantages of using database management system over traditional filing system?
3. Why indexing is essential in database? Differentiate dense index from sparse index with suitable example.
4. What is data abstraction? What are the three levels of data abstraction? Explain.
5. Define schedule and serializability. How can you test the serializability?
6. Describe roles and responsibilities of DBA.
7. Explain physical data independence and logical data independence.
8. What are the states of transaction model? Describe the ACID properties of transaction.

Group C

1*10=10

Attempt any ONE

9. Define normalization. Describe the importance of normalization in DBMS. Describe 1NF, 2NF, 3NF, BCNF and 4NF with examples.
 10. What is data model? What are its types? Explain the process of converting ER diagram to table.
 11. Define query processing and steps in query processing.
- Table Worker(Worker_id, Fname, Lname, salary, Joining_date, department)
- Table Bonus(Worker_ref_id, Bonus_date, Bonus_amt)
- Table Title(Worker_ref_id, Worker_title, Effective_from)
- a. Write an SQL query to print details of the Workers whose FIRST_NAME contains 'a'.
 - b. Write an SQL query to print details of the Workers whose SALARY lies between 100000 and 500000.
 - c. Write an SQL query to fetch the no. of workers for each department in descending order.
 - d. Write an SQL query to print details of the Workers whose bonus amount is more than 10000.



Tribhuvan University
Faculty of Humanities & Social Sciences
Ratna RajyaLaxmi Campus
Pradarshani Marga, Kathmandu
Pre-Board Examination 2026

Bachelor in Computer Application (BCA)
Course Title: Software Engineering
Code No: CACS253
Semester: IV

Full Marks: 40
Pass Marks: 16
Time: 2 hours

Candidates are required to answer the questions in their own words as far as possible.

Group – 'B'

Attempt any FOUR questions.

[4 x 5 = 20]

2. What are the differences between software engineering and system engineering? Explain.
3. What is a prototype model? Under what circumstances it is beneficial to construct a prototype model.
4. Why functional and non-functional requirements must be considered together while preparing software specification? List some of the functional requirements and non-functional requirements for "Online Library Management System" of a college.
5. Explain Repository model of architectural design with advantages and disadvantages.
6. What do you mean by coding standards? Why do we need to follow it? Explain the purpose.
7. What is regression testing? Why is regression testing necessary?
8. What is CMM? Describe its levels & compare it with ISO 9001.

Group – 'C'

Attempt any ONE Question.

[1 x 10 = 10]

9. What is requirements validation? What are the different types of requirements validation techniques? Explain.
10. Explain about the interface evaluation cycle in user interface design with a suitable figure.
11. Define maintenance. What are the types of software maintenance? Explain them in detail.

Best of Luck!!!